Musings

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1 The area of a circle

I know that number theory is strictly forbidden. People have wasted their whole lives on the subject.

I realised that I never once, in my whole university career, calculated the area of a circle. So I got Claude to do it.

The method is Archimedes. Inscribe a polygon in the circle. Double the number of sides. Each doubling halves the angle, and the half angle formula turns a square root into a nested square root. Starting from the octagon and doubling three times gives the area of the polygon with sixty four sides.

$$A_{64} = 64 \times \sqrt{\frac{1+\sqrt{\frac{1+\sqrt{\frac{1+\sqrt{\frac{1+\frac{\sqrt{2}}{2}}{2}}}{2}}}{2}}}{2}} \times \sqrt{\frac{1-\sqrt{\frac{1+\sqrt{\frac{1+\sqrt{\frac{1+\frac{\sqrt{2}}{2}}{2}}}{2}}}{2}}}{2}}$$

That is 3.136548, against $\pi = 3.141593$. Keep doubling and it converges.

And this raised a whole number of questions.

2 Is there a pattern

The formula is built out of nothing but square roots of whole numbers. So I wanted to know what a square root actually looks like. The honest answer is its continued fraction.

Every square root of a non square whole number repeats forever.

$\sqrt{2}$	$[1;\ \overline{2}]$
$\sqrt{3}$	$[1;\ \overline{1,2}]$
$\sqrt{5}$	$[2;\ \overline{4}]$
$\sqrt{6}$	$[2;\ \overline{2,4}]$
$\sqrt{7}$	$[2;\ \overline{1,1,1,4}]$
$\sqrt{13}$	$[3;\ \overline{1,1,1,1,6}]$
$\sqrt{19}$	$[4;\ \overline{2,1,3,1,2,8}]$
$\sqrt{31}$	$[5;\ \overline{1,1,3,5,3,1,1,10}]$
$\sqrt{94}$	$[9;\ \overline{1,2,3,1,1,5,1,8,1,5,1,1,3,2,1,18}]$

The notation $[a; b, c, d, \ldots]$ means the tower

$$a + \cfrac{1}{b + \cfrac{1}{c + \cfrac{1}{d + \cdots}}}$$

and the bar marks the part that repeats.

Turns out there is a pattern. There are two of them.

2.1 Every block is a palindrome followed by twice the first term

Take the repeating block and drop its last term. What is left reads the same backwards as forwards. The term you dropped is always twice the leading term.

$$\sqrt{19} = [\,4;\ \underbrace{\overline{2, 1, 3, 1, 2,}}_{\text{palindrome}}\ \overline{8}\,] \qquad \sqrt{31} = [\,5;\ \underbrace{\overline{1, 1, 3, 5, 3, 1, 1,}}_{\text{palindrome}}\ \overline{10}\,]$$

I had Claude check this for every non square below two thousand. There are no exceptions. It is a theorem of Galois and Legendre.

2.2 Numbers near a square have exact formulas

Five families. In each one k is the whole part of the root. The last two need $k \geq 2$.

n	expansion	examples
$k^2 + 1$	$[\,k;\ \overline{2k}\,]$	$2, 5, 10, 17, 26, 37, 50, 65$
$k^2 + 2$	$[\,k;\ \overline{k, 2k}\,]$	$3, 6, 11, 18, 27, 38, 51$
$k^2 + 2k$	$[\,k;\ \overline{1, 2k}\,]$	$3, 8, 15, 24, 35, 48, 63$
$k^2 + k$	$[\,k;\ \overline{2, 2k}\,]$	$6, 12, 20, 30, 42, 56, 72$
$k^2 + 2k - 1$	$[\,k;\ \overline{1, k-1, 1, 2k}\,]$	$7, 14, 23, 34, 47, 62, 79$

So $\sqrt{50}$ is $[\,7; \overline{14}\,]$ with no calculation at all, because fifty is seven squared plus one.

2.3 Where the pattern stops

The length of the repeating block is not predictable.

n	43	44	45	46	47	48	50	58	61	65	67
period	10	8	6	12	4	2	1	7	11	1	10

$\sqrt{61}$ needs eleven terms. $\sqrt{65}$ needs one. No formula gives that row.

This is not noise. It is a harder question wearing a disguise.

Cut the tower off after i steps and you get an ordinary fraction h/k. Now compute $h^2 - nk^2$ and watch it as i grows. Here is $\sqrt{13}$.

i	a_i	h/k	h^2-13k^2	
0	3	3/1	-4	
1	1	4/1	3	
2	1	7/2	-3	
3	1	11/3	4	
4	1	18/5	-1	end of the first block
5	6	119/33	4	
6	1	137/38	-3	
7	1	256/71	3	
8	1	393/109	-4	
9	1	649/180	1	end of the second block

The column hits ± 1 at the end of a block and nowhere else. I had Claude check that for every non square below three thousand and there are no exceptions. So the repeating block is precisely the stretch you have to walk to get from one of these to the next.

A block of even length lands on $+1$ at the end of the first block. A block of odd length lands on -1 there, so you must walk a second block to reach $+1$.

$$\text{steps to the smallest solution of } x^2 - ny^2 = 1 \;=\; \begin{cases} L-1 & L \text{ even} \\ 2L-1 & L \text{ odd} \end{cases}$$

So the period tells you how many steps you must take. It does not tell you how big the answer is. That comes from the terms you walk past, because each step multiplies the numerator by roughly a_i.

n	period	smallest x, y
2	1	3, 2
13	5	649, 180
29	5	9801, 1820
61	11	1766319049, 226153980
109	15	158070671986249, 15140424455100

Fermat set $n = 61$ as a challenge to the English in 1657. He picked it because the block is long and the answer is enormous. Bhaskara the Second had solved it five hundred years earlier.

Thirteen and twenty nine both have period five, so both need nine steps. The terms along the way differ, so the answers differ by a factor of fifteen. The length says how far. The terms say how big.

Greater minds than myself have pondered these questions. But now with Claude a mediocre mathematician such as myself can consider them.

3 Sturmian sequences

Here is the second question. An irrational number is an infinite object. What does it look like as a sequence of symbols.

Take an infinite sequence of 0s and 1s.

$$\ell \;=\; 0\,1\,0\,0\,1\,0\,1\,0\,0\,1\,0\,0\,1\,0\,1\,0\cdots$$

Slide a window m letters wide along it and write down every distinct block you see through it. Let $F(m)$ be how many distinct blocks there are.

Do it. Take $m = 3$ and walk the window along the sixteen letters above.

position	window	block	
0	010**010**1001001010	010	new
1	0**100**101001001010	100	new
2	01**001**01001001010	001	new
3	010**010**1001001010	010	seen
4	0100**101**001001010	101	new
5	01001**010**01001010	010	seen
6	010010**100**1001010	100	seen
7	0100101**001**001010	001	seen
8	01001010**010**01010	010	seen
$\cdots$ and nothing new appears after that $\cdots$			

Four distinct blocks, so $F(3) = 4$. Four of the eight possible three letter blocks never appear at all. This sequence is far from arbitrary.

Now widen the window by one letter. For each of the four blocks, collect every letter seen following it anywhere in the sequence.

block	occurs at	followed by	becomes
010	$0, 3, 5, 8, 11, 13$	0 at 0, **1** at 3	0100 *and* 0101
100	$1, 6, 9$	1, every time	1001
001	$2, 7, 10$	0, every time	0010
101	$4, 12$	0, every time	1010

Three of the four pass through unchanged. One does not. 010 is followed by a 0 at position 0 and by a 1 at position 3, so it becomes two blocks. Say that it *splits*. So $F(4) = 5$.

Here is why the count is worth having. Suppose you are reading ℓ one letter at a time, trying to call the next letter, and you can remember only the last m. Two occurrences of the same block are then indistinguishable to you. If that block splits, the next letter genuinely differs between the two occurrences, so you are wrong at one of them. Not through bad judgement. The letters you can see do not contain the answer.

Definition 1. A sequence is *Sturmian* when exactly one block of each length splits.

$$F(m) = m + 1 \qquad \text{for every } m \geq 1.$$

So a Sturmian sequence is one where, at every width, there is precisely one situation your memory cannot resolve. More memory does not remove it. It only makes it rarer.

4 One sequence for each irrational

Theorem 2 (Morse and Hedlund)**.** *A sequence ℓ is Sturmian if and only if there are an irrational $\alpha \in (0, 1)$ and a $\rho \in [0, 1)$ with*

$$\ell_k \;=\; \lfloor (k+1)\alpha + \rho \rfloor - \lfloor k\alpha + \rho \rfloor \qquad \textit{for all } k \geq 0.$$

We do not prove this. The slope α is the density of 1s and is determined by ℓ. The intercept ρ only says where you started reading.

So there is exactly one Sturmian sequence for each irrational number, up to a shift. The sequence above has slope $1/\varphi^2$ and is the Fibonacci sequence, the fixed point of $0 \mapsto 01$ and $1 \mapsto 0$.

That is a bijection between two things I did not expect to meet. On one side the irrational numbers. On the other side the symbol sequences that are minimally unpredictable.

5 Now take the irrational to be a square root

This is the question I actually wanted to ask.

The theorem hands me a sequence for every irrational. Section 2 tells me that square roots are the irrationals whose continued fractions repeat. So what is special about the sequence when $\alpha = \sqrt{n} - \lfloor \sqrt{n} \rfloor$.

There is a renormalisation here. Take a Sturmian sequence and pass to its derived sequence, the one recorded only at the sites where the shorter window could not decide. The derived sequence is Sturmian again, and its slope is

$$G(\alpha) \;=\; \left\{ \tfrac{1}{\alpha} \right\}.$$

This is the Gauss map. On a continued fraction it does exactly one thing. It deletes the first term.

$$[0;\, a_1, a_2, a_3, \ldots] \;\longmapsto\; [0;\, a_2, a_3, a_4, \ldots]$$

So the tower of derived sequences is the continued fraction, read one term at a time. For a generic irrational that list never repeats.

For a square root it does.

slope	orbit under the Gauss map	cycle
$\sqrt{2}-1$	$0.414214 \to 0.414214 \to \cdots$	1
$\sqrt{3}-1$	$0.732051 \to 0.366025 \to 0.732051 \to \cdots$	2
$\sqrt{5}-2$	$0.236068 \to 0.236068 \to \cdots$	1
$\sqrt{6}-2$	$0.449490 \to 0.224745 \to 0.449490 \to \cdots$	2
$\sqrt{7}-2$	$0.645751 \to 0.548584 \to 0.822876 \to 0.215250 \to 0.645751$	4
$\sqrt{8}-2$	$0.828427 \to 0.207107 \to 0.828427 \to \cdots$	2

The cycle length is the period of the continued fraction. It is the column from Section 2, the one I said had no formula, now reappearing as the depth at which a sequence starts repeating itself.

So the square root sequences are the self similar ones. The splitting never stops, so no finite table of remembered windows predicts them. But the renormalisation closes after finitely many levels, so a finite recursion does.

That gap is a real place. A periodic sequence is solved by a table. A generic Sturmian sequence is solved by nothing finite. The square root sequences sit between the two, and the period of the continued fraction is how far in they sit.

There is a theorem here which I state and do not prove. It is due to Crisp, Moran, Pollington and Shallit.

Theorem 3. *A Sturmian sequence is the fixed point of a finite substitution if and only if its slope is a quadratic irrational.*

Which says that the square root sequences are exactly the Sturmian sequences a finite machine can generate.

6 What I take from this

Three things, none of which I would have reached alone.

The first is that the area of a circle is a question about square roots, and a square root is a repeating pattern with a palindrome in it.

The second is that the length of that pattern is not a curiosity. It is the size of the smallest solution of a Pell equation, which is why nobody has a formula for it.

The third is that the same number appears twice, in two settings that have no business being related. Once as the period of a continued fraction. Once as the depth at which a symbol sequence becomes self similar.

I do not know whether any of this is new. It is almost certainly not. But I found it by asking, and the asking took an afternoon rather than a career, which is the part that seems worth reporting.